Independent projectile directions

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Learning target: Explain why horizontal and vertical projectile motion share time but evolve differently.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 6.*
g=5 N/kg; force 35 N toward the source.

02. Worked example: explain the model
Horizontal arrow stays 3 m/s; vertical arrow begins zero and grows to 20 m/s downward.

03. Read and interpret the model
Same fall time if same initial vertical velocity and negligible air resistance.

04. Apply the physics
6 m; range doubles to 12 m, fall time unchanged.

05. Explain your reasoning
Horizontal velocity alone does not change vertical acceleration; lift/another vertical interaction would be required, outside this ideal model.

06. Review: connect an earlier idea *Review: Unit 1, Day 19.*
The student multiplied by mass. Divide: $a=F_{net}/m=6\ m/s^2$. Check units $N/kg=m/s^2$ or substitute ma back into F_{net}.

07. Exit check
Equal times; second range 3 times first.

08. STRETCH: test the model
Same fall time; ranges 6 and 15 m. Faster horizontal motion does not delay falling in the ideal model.

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Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 6.*
Field 3 N/kg; force 2.1 N toward the source.

02. Worked example: explain the model
Horizontal arrow stays 3 m/s; vertical arrow begins zero and grows to 20 m/s downward.

03. Read and interpret the model
Same fall time if same initial vertical velocity and negligible air resistance.

04. Apply the physics
Δx 12 m;Δy=-11.25 m; v=(8,-15) m/s.

05. Explain your reasoning
Horizontal velocity alone does not change vertical acceleration; lift/another vertical interaction would be required, outside this ideal model.

06. Review: connect an earlier idea *Review: Unit 1, Day 19.*
Central 52 m; half-width 5 m; interval [47, 57] m. Valid checks include units and upper/lower endpoint substitution.

07. Exit check
Speed 10 m/s; need initial position and elapsed motion/history.

08. STRETCH: test the model
(6,-2) m/s; horizontal remains constant without horizontal force. Both motions share elapsed time.